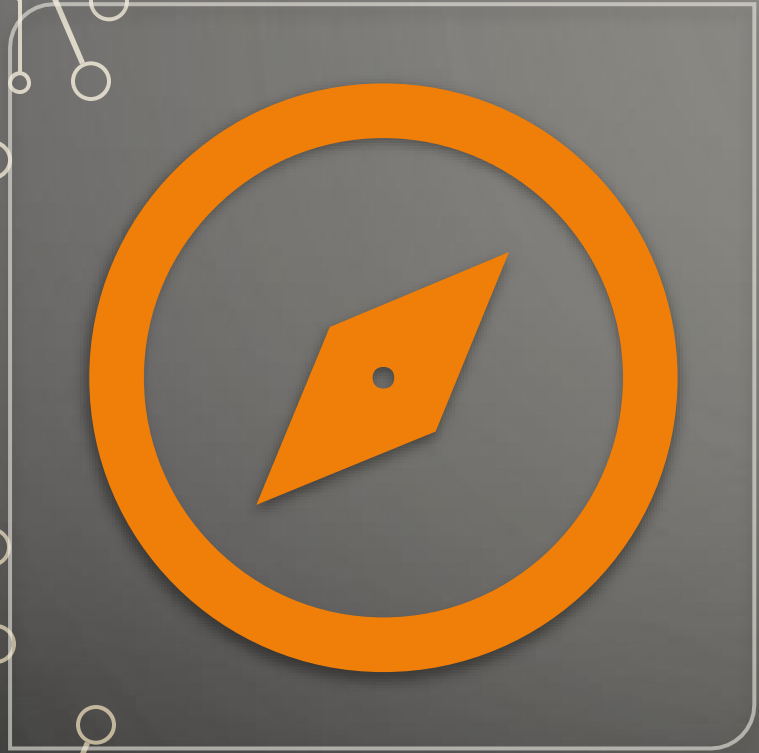




**ORACLE DATABASE:
EVOLVING FROM ENGINE
TO INTELLIGENCE**

WHO AM I?



Engineer and Advocate at **Redgate**

Keynote & technical speaker, podcast contributor

27 Years in the Tech Industry

DBA (Oracle, SQL Server, MySQL, PostgreSQL...)

Author of numerous books, articles and blogs

DevOps, Analytics and AI

Cloud Migrations and High IO Infrastructure

Co-Leader for the MS Data Platform DEI Group

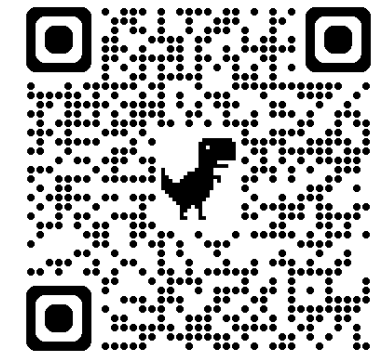
Organizer for the PDX WIT Group



<https://www.linkedin.com/in/kellyngorman/>



<https://dbakevlar.com>



ORACLE HAS
BEEN GOOD
TO ME

1997- “Make a DBA out of her.”

2000- Met Tim, as a mentor.

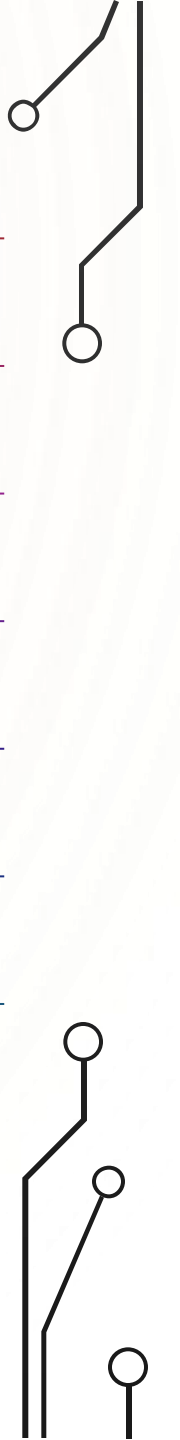
2010- Blogging, speaking and Tim re-entered my life as more.

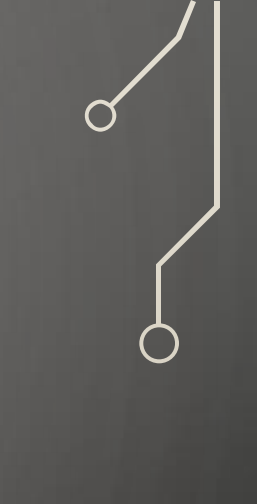
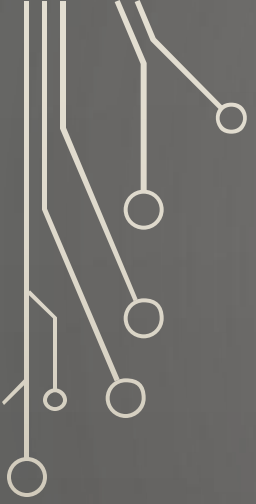
2014- Worked for Oracle

2018- Worked for Microsoft....doing Oracle

5 kids through college, launced, traveled the world

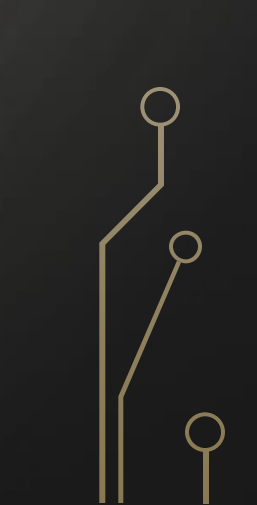
Our lives would not be where they are without Oracle.





AS WITH MANY TECHNOLOGISTS, I'VE LEFT FOR NEW...

- Oracle was the database engine I've always been in the deepest with technically.
- Infrastructure focus- Oracle was often the heavy weight requiring my expertise.
- AI- It all comes back to data....so Oracle returned.



Oracle has incredible career gravity...

THE LEGACY WE STAND ON

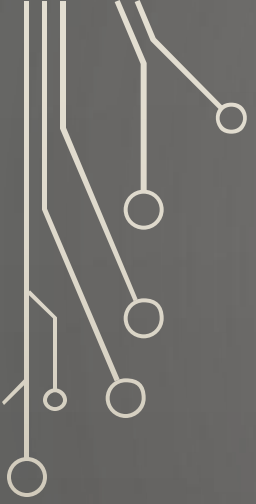
Oracle Database has 40+ years powering mission-critical systems

Industry benchmark for performance, resilience, availability

Trusted for financials, healthcare, government, global-scale apps

We've *mastered* data management-
Now **DATA INTELLIGENCE**





PREMIUM FEATURES OF ORACLE

- Architecture: Single Instance, Multitenant, RAC, Data Guard, Exadata
 - MVCC: Oracle's UNDO model, PIT consistent reads, etc.
 - The Cost-Based Optimizer
 - Multi-model Database Leadership
 - DBMS Packages in PL/SQL
 - Mature Tooling Ecosystem: RMAN, Data Pump, GG, etc.
- 
- 

WHAT INSPIRED ME ABOUT ORACLE?

Performance data-

- UTLBSTAT and UTLESTAT
- Statspack
- Advanced tracing
- AWR, ASH and ADDM

Very Large Databases (VLDBs)

- Data warehouse design
- Partitioning, Mviews
- Exadata
- PL/SQL

WE'VE GONE FROM



ORACLE 8



ORACLE
9i(INTERNET)



ORACLE
10g(RID)



ORACLE
12c(CLOUD)



ORACLE 23
AND 26AI

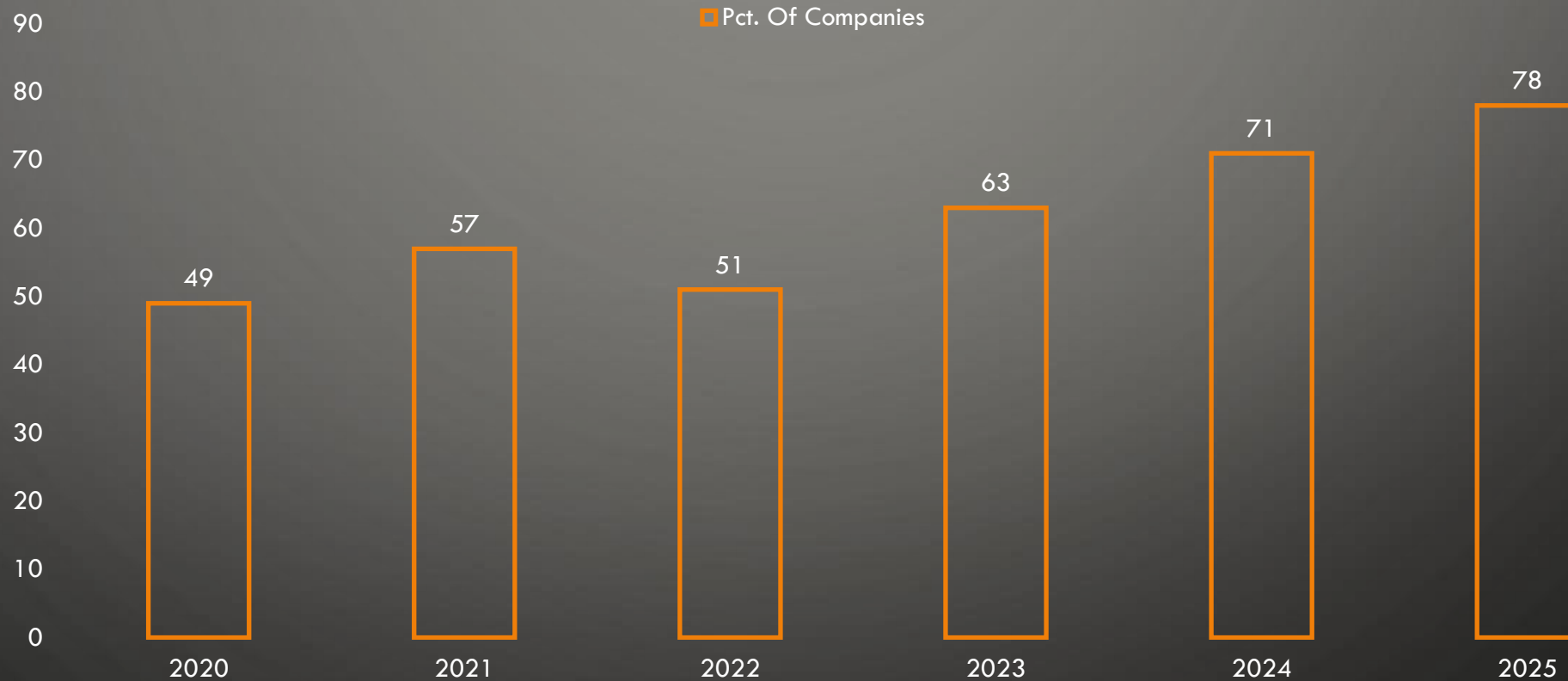
YEAH, I SAID
“AI”...



AI IN ENTERPRISE

- AI/ML is in production

Percentage Of Companies



ADD YOUR ORACLE VOICE TO THE STATE OF DATABASE LANDSCAPE!

- Join 1000's of contributions
- Provide more Oracle user info
- Download the report to see the trends

Save me from frustration! 😊





SO WHY IS ORACLE CHANGING NOW?

- AI is rewriting expectations for every workflow
 - Data volumes are exploding
 - Automation is no longer optional
 - Organizations demand:
 - Real-time insight
 - Predictive capability
 - Governed, trustworthy data
- 

Data professionals must shift to intelligence enablers.

SYSTEM OF RECORD IS THE SYSTEM OF REASONING

Yesterday

- Store data
- Protect data
- Deliver performance

Today/Tomorrow

- Analyze in place
- Reason with data
- Govern automatically
- Augment workflows with AI

Oracle 26ai is the bridge.

ORACLE 26AI PLATFORM

Three Core Pillars:

- Autonomous Automation
- AI + ML Inside the Database
- Governed, Trusted Data at Scale



Oracle is not adding more features but redefining the database. The shift is absolutely felt.

YOU ASK- WHY PUT AI IN THE DATABASE?

Data is at risk when data moves



Security & governance weaken outside of the database



Performance hits from ETL to external models and back



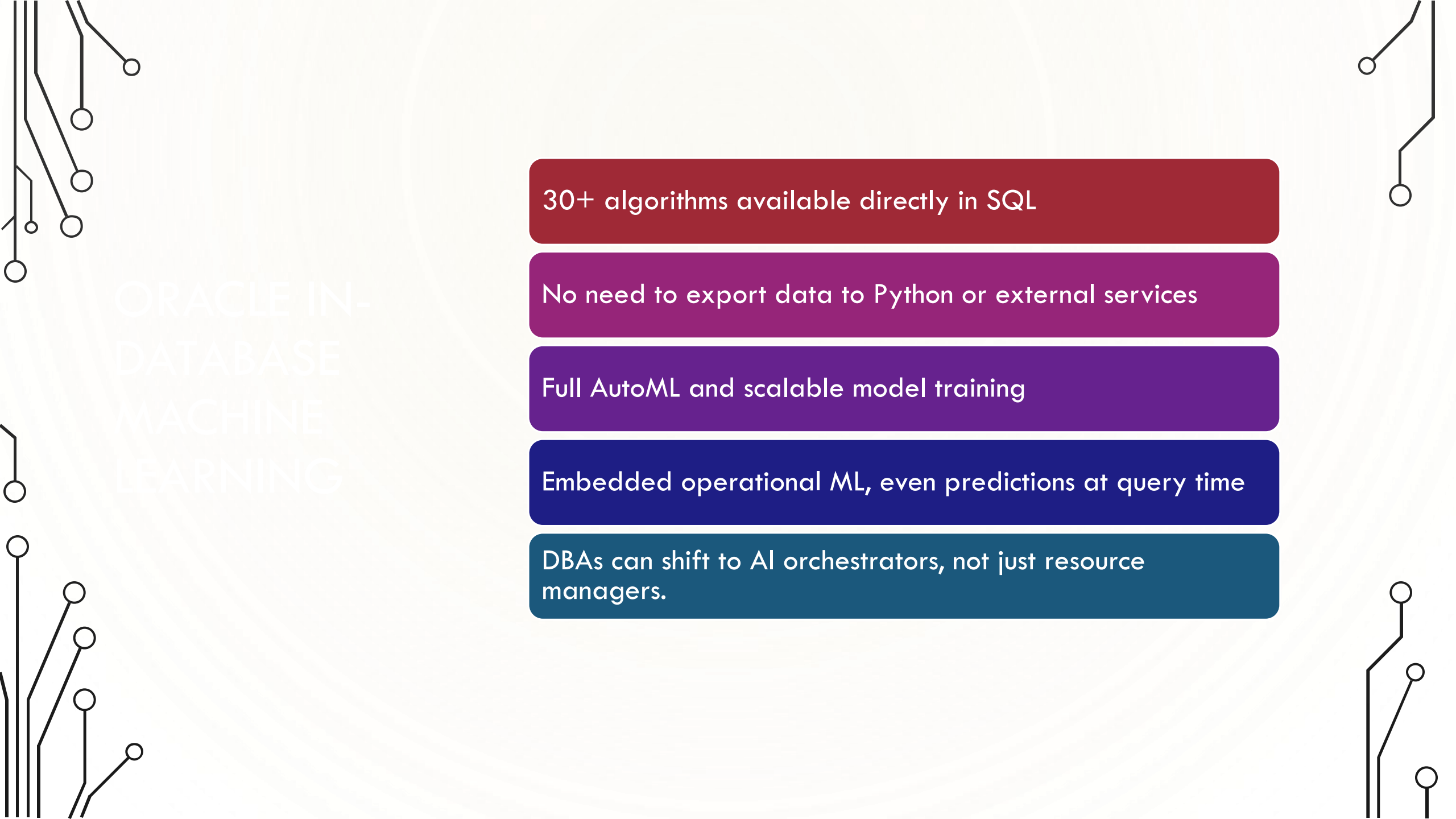
In-database AI makes sense:

Safer

Cheaper

More secure

More accurate



ORACLE IN- DATABASE MACHINE LEARNING

30+ algorithms available directly in SQL

No need to export data to Python or external services

Full AutoML and scalable model training

Embedded operational ML, even predictions at query time

DBAs can shift to AI orchestrators, not just resource managers.

VECTOR SEARCH, AI'S NEW INDEX

Native vector data types

Indexing for embedding & semantic search

Ideal for:

- RAG architectures
- Document intelligence
- Anomaly detection
- Personalize recommending

Enables AI reasoning inside the database.

JSON DUALITY WITH MODERN DATA MODELS



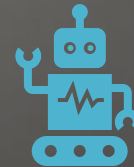
Oracle's JSON and
relational worlds
coverage



Dual representation
for both structured
and unstructured



Developers can
build modern apps
without losing
governance



Simplified
microservices & AI-
driven API
ecosystems



ORACLE DATABASE
AND AI ARE
THE PERFECT
COLLABORATION

Relational AI vs. Shadow AI

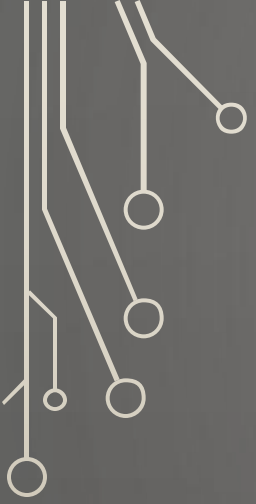
- 11% of ChatGPT free prompts include PII.
- 13% of chatbot entries contained PII.

<https://www.cyberhaven.com/blog/4-2-of-workers-have-pasted-company-data-into-chatgpt>

Uploaded Files to AI – Even Worse

- 40% of uploaded files to AI included PII.

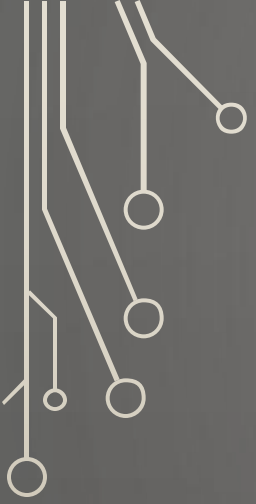
<https://www.esecurityplanet.com/news/shadow-ai-chatgpt-dlp/>



Oracle 26ai is Data-Aware

Understands complex Oracle architectures

- from multitenant to RAC
 - adapts to your environment.
- 
- 



This Enhancement Accelerates Root Cause Analysis

- AI is good at the messy and Oracle is complex.
 - A way to navigate Oracle documentation from a single location, internal and external is powerful.
- 
- 



Enhances (Not Replaces) Human Insight



Offers suggestions but leaves critical decisions to experienced professionals who understand business context.



The image features a dark gray background with a subtle, large-scale grid pattern. In the four corners, there are decorative elements resembling circuit board traces or neural network connections, consisting of thin white lines and small white circles.

*AI doesn't replace DBAs and
Developers - it completes them.*

DEVELOPMENT IN ORACLE

Direct joins in UPDATE/DELETE statements

Aliases in GROUP BY/HAVING

Table-value constructors

Annotations/Metadata

DBMS* internal packages

Smarter Performance Tuning

- **SQL tuning suggestions**
Analyzes execution plans and offers indexing or rewriting tips.
- **Pattern recognition**
Detects performance trends from AWR, ASH, and v\$ views faster than manual review.

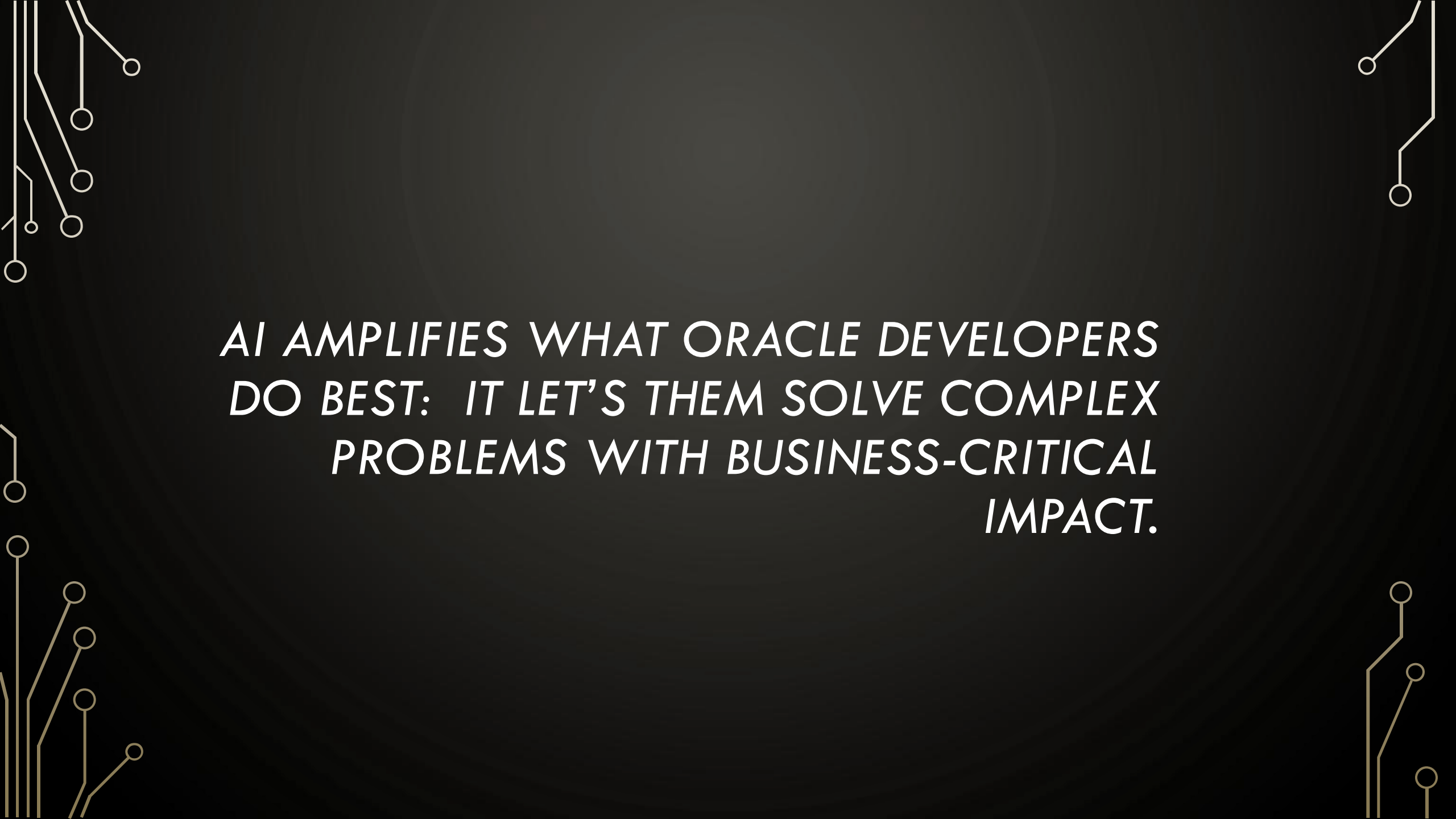
Improved Troubleshooting

- **Explain plan analysis**
Breaks down plans in plain language, highlighting inefficient steps.
- **Root cause insight**
Correlates logs, wait events, and session info to pinpoint bottlenecks.

Instant Oracle knowledge base
Summarizes documentation, error codes, and usage examples in seconds.

Code explanation
Makes legacy or poorly documented code understandable.



The image features a dark gray background with a subtle, large-scale grid pattern. In the four corners, there are decorative elements resembling circuit board traces or neural network connections, consisting of thin white lines and small circles.

*AI AMPLIFIES WHAT ORACLE DEVELOPERS
DO BEST: IT LET'S THEM SOLVE COMPLEX
PROBLEMS WITH BUSINESS-CRITICAL
IMPACT.*

The image features a dark gray background with white, stylized circuit board traces in the corners. These traces form various geometric shapes and end in small circles, resembling electronic components or connectors. The traces are located in the top-left, top-right, bottom-left, and bottom-right corners, framing the central text.

HIGH AVAILABILITY AND REPLICATION SECOND TO NONE

HA IS EVOLVING AGAIN



Demands for real-time access are accelerating



Hybrid-cloud and multi-cloud architectures are standard



AI-driven workloads change the world of uptime & continuity



Oracle 23ai and 26ai introduced next-level HA features

RAFT REPLICATION



MODERN
CONSENSUS-
BASED
REPLICATION



FASTER, MORE
PREDICTABLE
FAILOVER



REDUCES DATA
LOSS (NEAR-
ZERO RPO)



IDEAL FOR
DISTRIBUTED,
HYBRID AND
CLOUD
DEPLOYMENTS



DESIGNED FOR
REAL-TIME AND
AI WORKLOADS
IN MIND

READ-ONLY HYBRID PDBS



New flexibility for global access patterns



Great fit for analytics, AI model scoring and read scaling



Lower cost, but higher resiliency



Continues Oracle's direction towards containerization

The image features a dark gray background with a subtle, large-scale pattern of concentric circles. In the four corners, there are decorative elements resembling circuit board traces or data lines. These lines are white in the top-left and top-right corners, and a light beige or gold color in the bottom-left and bottom-right corners. Each corner design consists of several lines of varying lengths and angles, some terminating in small open circles.

DATA TRUST AT SCALE

THE TRUST PROBLEM

AI is only as good as
the data that feeds it.

If we can't trust the
data, we can't trust
the model.



IMMUTABLE BLOCKCHAIN TABLES

Cryptographically chained rows

Prevent tampering or silent modification

Perfect for:

- Compliance
- Financial systems
- Regulatory environments
- High-trust audit trails

Trust is no longer an afterthought





UNIFIED AUDIT & POLICY- DRIVEN GOVERNANCE

One audit framework across Oracle

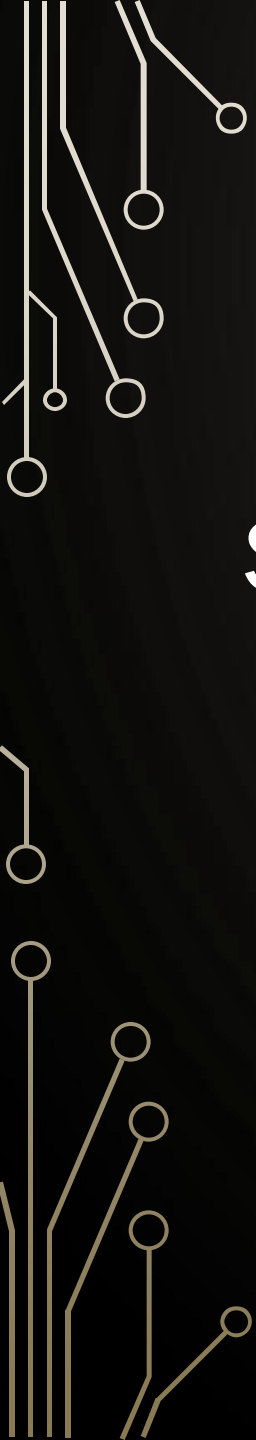
Fine-grained, unified policies

Automated controls for:

- GDPR
- HIPAA
- SOX
- Internal data governance

Keeps humans focused on stewardship, not policing



A decorative graphic of white circuit lines with circular nodes, resembling a stylized network or data flow, is positioned on the left side of the slide.

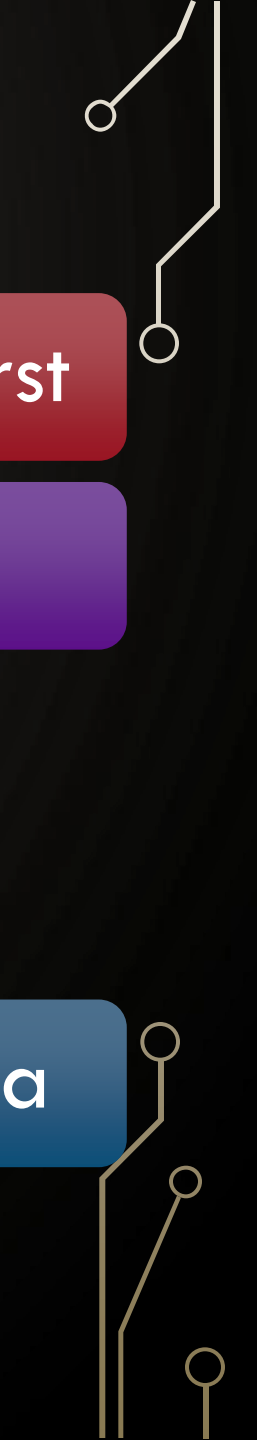
SECURITY IS THE NEW COOL

Adversarial attacks target data first

New Features include:

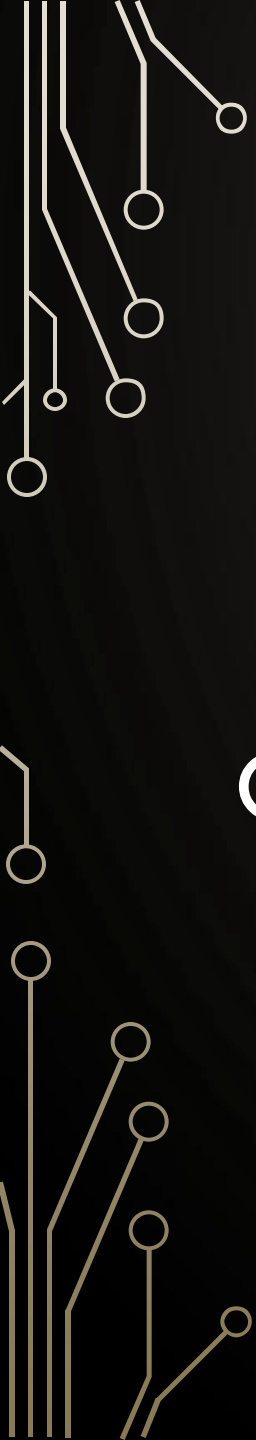
- Encryption everywhere
- Policy-driven access
- Data masking & redaction
- Row-level and label security

Aligns with the new reality of data

A decorative graphic of white circuit lines with circular nodes, resembling a stylized network or data flow, is positioned on the right side of the slide.

The image features a dark gray background with white, stylized circuit board traces in the corners. These traces form a network of lines and small circles, resembling electronic components or data pathways. The traces are located in the top-left, top-right, bottom-left, and bottom-right corners, framing the central text.

THE HUMAN FACTOR



ROLES ARE CHANGING

Old Roles

DBAs — Performance, patching,
backups

Developers — Development code,
app logic

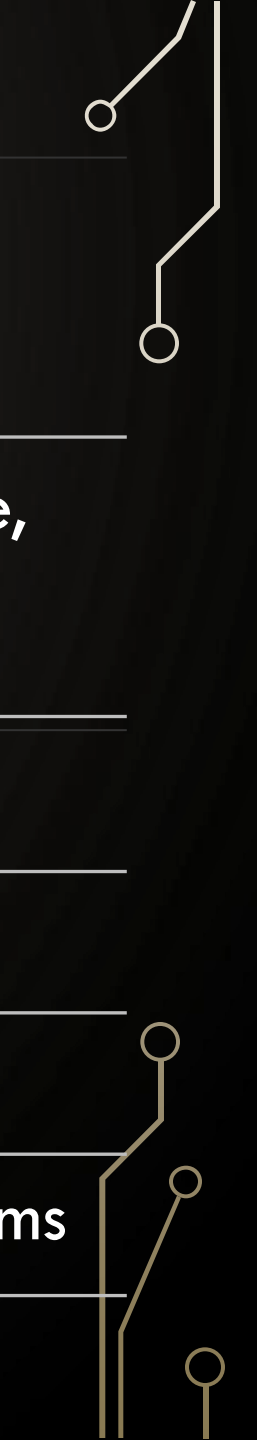
New Roles

Data stewards

AI orchestrators

Governance enablers

Architects of intelligent ecosystems





WHAT THIS MEANS FOR DBAS

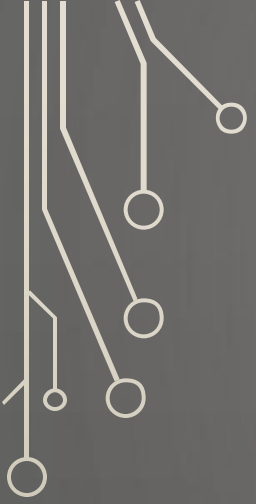
Less time patching, more time optimizing

Less firefighting, more architecture

We will grow into:

- AI workflow architect
- Data product manager
- Governance strategist

Your experience becomes even more valuable in the AI era.

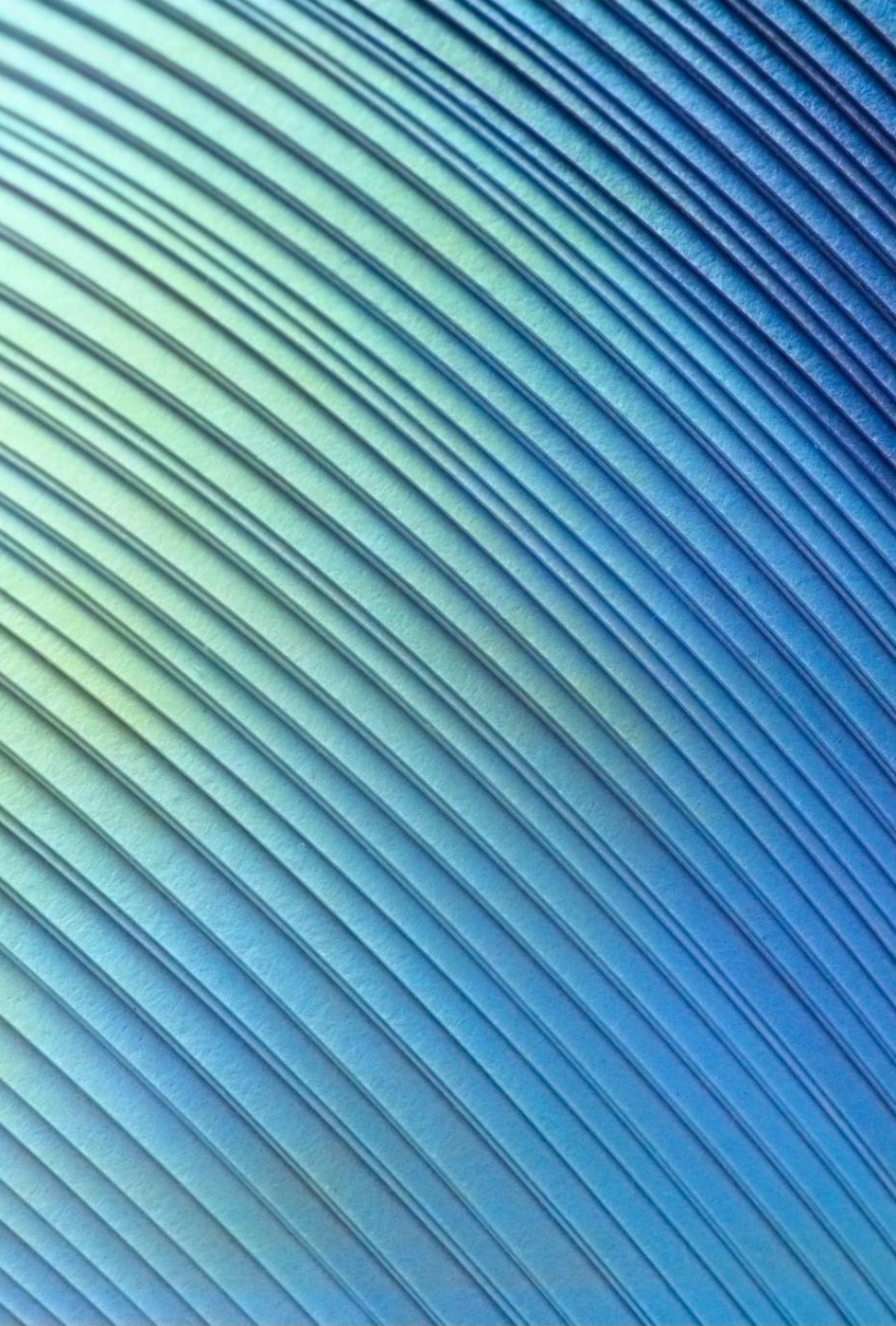


TIMELESS TRUTHS

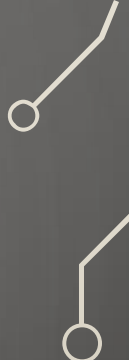
- No matter how much AI changes
 - Data still requires protection
 - Workloads still require optimization
 - Organizations still require integrity

AI accelerates everything, but it does not ***replace accountability.***





WHAT THIS MEANS FOR DEVELOPERS



Build intelligent apps without managing ML infrastructure

Leverage JSON duality, vector search and ML in SQL

More focus on innovation, less on integration

New Patterns develop: AI-driven application design.



WHAT THIS MEANS FOR ARCHITECTS

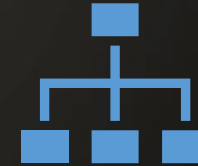


**Architect for reasoning, not
just infrastructure**

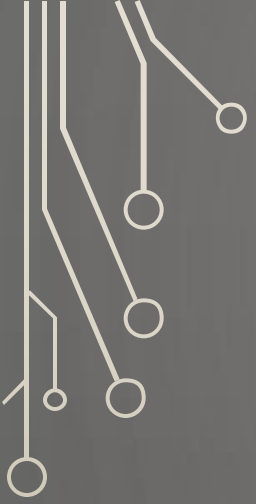


New patterns:

In-database AI
Hybrid PDBs
RAFT-based HA topologies



**Data governance becomes a
priority.**



FROM INSIGHT TO INTELLIGENCE

We're moving from managing data to
making *data intelligent*

And you are the center of this transformation.



INFRASTRUCTURE SKILLS

Data Storage & Engineering Tools

Cloud infrastructure

- Virtual Machines
- Storage
- Network
- Identity/authentication

Cloud services

Docker/Kubernetes

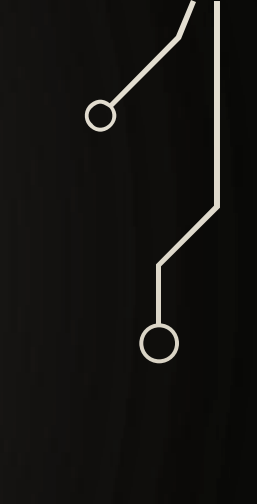
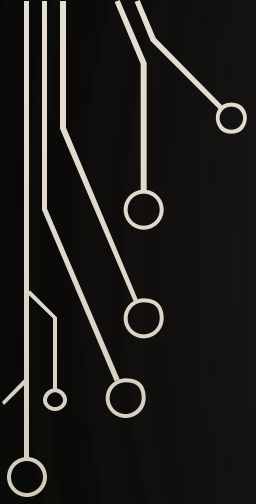


THE FUTURE OF AI JOBS ARE PROMISING AND MULTI-FACETED:

- 64% increase in roles in Data analysts
- 180% increase in Data Scientists or specialization in data science.
 - AI Model Trainers: Experts who fine-tune models
 - Data Engineers: Prepare data for study
 - Machine Learning Engineers: Design and implement ML apps
- Data Science Ethics Roles will be created.
Actual data analysis will become more automated.



**THE WORLD ECONOMICS FORUM
PROJECTS AN INCREASE OF 170
MILLION NEW JOBS CREATED WHILE
THERE ARE 92 MILLION DISPLACED
DUE TO AI GLOBALLY BY THE YEAR
2030.**



YOU REMAIN THE CONSCIENCE OF THE DATA

AI can reason

AI can automate

AI can predict

But only people can ensure:

- True ethics
- Alignment
- Trust

Oracle empowers you to be that steward of responsible intelligence.



CALL TO ACTION THIS WEEK:

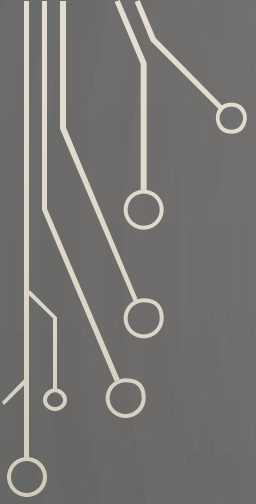
Explore new capabilities in Oracle 26ai

Attend conference sessions the next two days.

Network with those around you in sessions, gathering, etc.

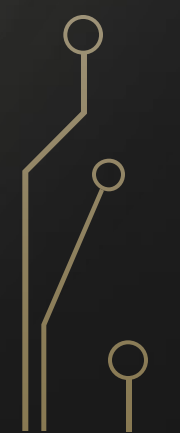
Lead the evolution toward intelligent data ecosystems.

You are not being replaced but amplified by the changes within Oracle.



MY HOPES FOR YOU

I hope Oracle:

- Removes the busy work and provides you more opportunities for career growth.
 - Provides you with more personalized learning- no matter how you learn.
 - Creates more convenience, safety, creativity and extensibility.
 - Builds more success for your organizations.
 - Brings you as much career satisfaction as it has for me.
- 
- 

THANK YOU!

Kellyn Gorman,

Engineer and Advocate at

Redgate

<https://www.linkedin.com/in/kellyngorman/>

<https://dbakevlar.com>

